

Winding-Free Exact Diagonalization of Quantum Spin Ice

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Quantum spin ice realizes an emergent three-dimensional $U(1)$ gauge theory whose low-energy dynamics is generated by six-spin ring exchange around pyrochlore hexagons. The sign of this exchange selects zero- or π -flux gauge backgrounds, yet the frustrated sign relevant to π -flux candidates is inaccessible to sign-problem-free quantum Monte Carlo. Exact diagonalization is sign unrestricted, but standard periodic clusters admit a noncontractible four-spin process at second order, parametrically above the physical third-order hexagon tunneling. We eliminate this finite-torus artifact by coupling microscopic spin transfers to a source that records their transport on the covering lattice and projecting the exact ice-descended band onto its zero-transport Fourier component. On cubic-16, the $M = 3$ and $M = 4$ projected operators agree within 0.013% in centered operator norm, and a third-order path classification removes every winding four-loop while retaining every contractible hexagon. At $(J_{\pm}, J_{\pm\pm})/J_{zz} = (0.046, 0)$, the low-temperature heat-capacity peak shifts from $0.02554J_{zz}$ in ordinary exact diagonalization to $0.001463J_{zz}$, close to the quantum Monte Carlo value $0.001100J_{zz}$. The longitudinal X -point excitation centroid simultaneously shifts from $0.08069J_{zz}$ to $0.005362J_{zz}$. The residual thermodynamic discrepancies isolate finite-size effects beyond the removed winding process.

Introduction.—Quantum spin ice (QSI) is a $U(1)$ quantum spin liquid with fractionalized excitations and an emergent gauge field whose gapless photon mode provides a characteristic low-energy signature [1–3]. Recent experiments on $\text{Ce}_2(\text{Zr}, \text{Sn}, \text{Hf})_2\text{O}_7$ suggest that Ce-based pyrochlores may realize a π -flux QSI, in which an emergent gauge flux of π threads each elementary hexagonal plaquette [Fig. 1(i.a)] [4–6]. In $\text{Ce}_2\text{Zr}_2\text{O}_7$, no long-range order or spin freezing is observed down to the lowest accessible temperatures, whereas recent heat-capacity measurements on $\text{Ce}_2\text{Hf}_2\text{O}_7$ resolve two anomalies near 65 mK and 25 mK [7]. Numerical linked-cluster fits place the candidate parameters within the π -flux QSI regime, motivating a microscopic explanation of this two-scale thermodynamic structure.

The minimal microscopic setting for QSI is the nearest-neighbor pseudospin-1/2 XYZ model

$$H_0 = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+) + J_{\pm\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^+ + S_i^- S_j^-), \quad (1)$$

with $J_{zz} > 0$. For $|J_{\pm}|, |J_{\pm\pm}| \ll J_{zz}$ the low-energy states obey the spin-ice rule $Q_t \equiv \sum_{i \in t} S_i^z = 0$ on every tetrahedron t ; we write P_{ice} for the orthogonal projector onto that manifold, and call a tetrahedron with $Q_t \neq 0$ a spinon. Transverse exchange first connects distinct ice configurations at third order, through collective flips around elementary six-site hexagons, generating the ring-exchange scale

$$g_6 = \frac{12|J_{\pm}|^3}{J_{zz}^2}. \quad (2)$$

In the convention of Eq. (1), $J_{\pm} > 0$ gives the zero-flux QSI, whereas $J_{\pm} < 0$ gives the π -flux QSI. The paramet-

rically small scale g_6 is therefore a natural candidate for the lower-temperature thermodynamic anomaly.

Obtaining a microscopic description of this regime using unbiased numerical methods remains challenging. Quantum Monte Carlo (QMC) simulations access the zero-flux QSI for $J_{\pm} > 0$ [8–10] but encounter a sign problem in the π -flux regime, $J_{\pm} < 0$. Numerical linked-cluster expansions fail to converge at the ultralow temperatures at which the lower anomaly emerges. Finite-temperature tensor-network calculations have reached a periodic 48-site three-dimensional Heisenberg cluster, but only well above the ring-exchange scale [11]. Density-matrix renormalization group and minimally entangled typical thermal states reach lower temperatures on pyrochlore tubes [12], but their quasi-one-dimensional geometry cannot support a gapless photon mode. A sign-unrestricted microscopic treatment that resolves the three-dimensional gauge dynamics at temperatures of order g_6 is therefore needed.

Exact diagonalization (ED) is exact on a finite cluster and imposes no sign restriction, but is limited to small pyrochlore tori, notably the cubic 16-site and face-centered-cubic 32-site clusters. Their dominant artifact is not size alone but the appearance of spurious four-site loops. On cubic-16, a noncontractible four-spin path closes through the periodic boundary [Figs. 1(i.b) and 1(i.c)] and contributes already at second order, with

$$g_4 = \frac{4J_{\pm}^2}{J_{zz}}, \quad \frac{g_4}{g_6} = \frac{J_{zz}}{3|J_{\pm}|}, \quad (3)$$

whereas the physical hexagon process appears only at third order, at the scale g_6 . The artifact therefore diverges relative to the physical process in the nominally controlled limit $|J_{\pm}|/J_{zz} \rightarrow 0$. Cubic-16 hosts 36 such winding four-cycles and 16 contractible hexagons; FCC-32 reduces but does not remove the problem, with 24

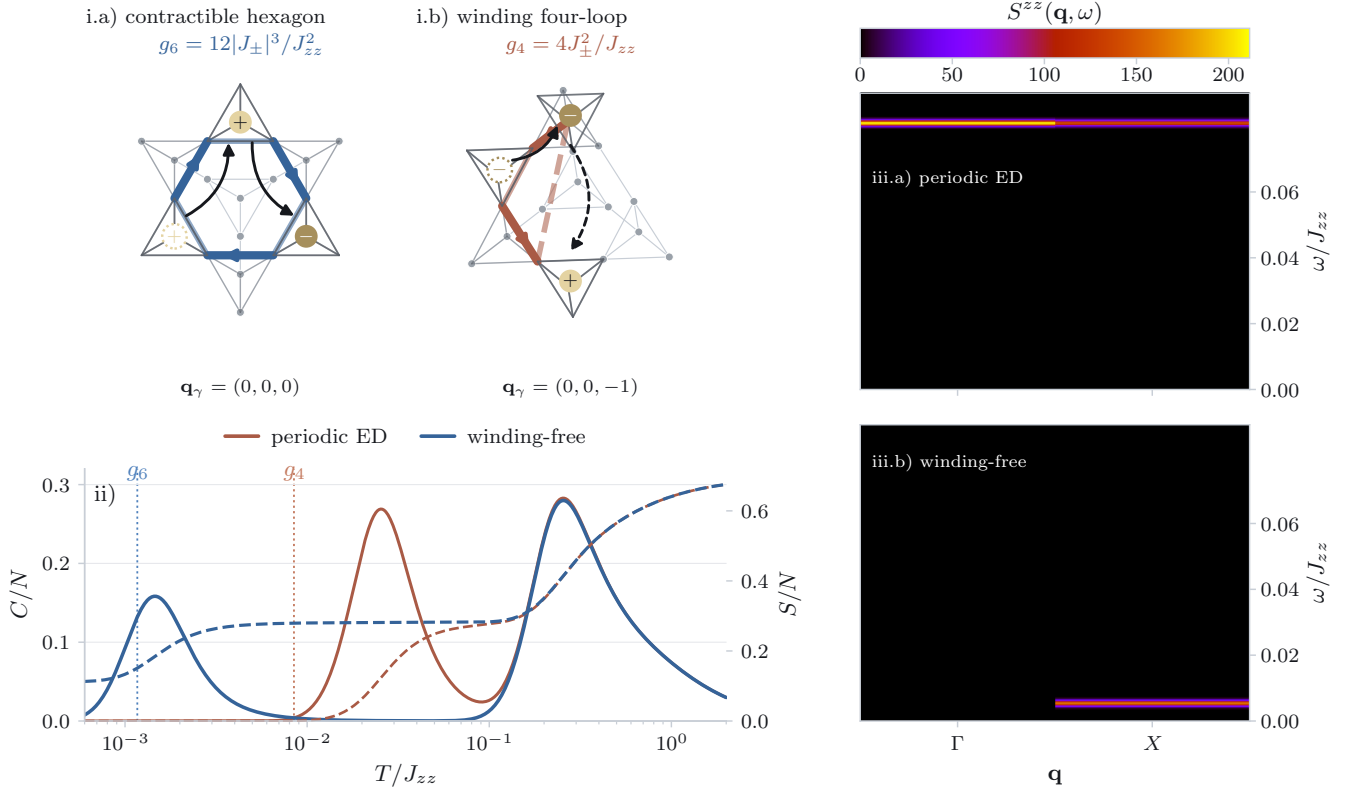


FIG. 1. Microscopic transport and winding removal on cubic-16. (i) The two shortest ice-to-ice processes on the 16-site torus. Gray marks the cluster; colored circles and bonds are the loop, with its exchanges numbered in order and each drawn as an arrow from the lowered to the raised spin. An exchange leaves the tetrahedron it straddles neutral and charges the two outer ones, so it creates, moves, or annihilates a spinon–antispinon pair; charges sit at the centre of their tetrahedron, whose frame is drawn, light for $Q_t > 0$ and dark for $Q_t < 0$, faded where a charge has since left. (i.a) On the contractible hexagon the pair is created, walks, and annihilates inside one cell, transporting nothing. (i.b) On the winding four-loop the second exchange carries a charge onto a copy of its tetrahedron that the loop reaches only across the periodic boundary (dashed bond); the pair closes solely through the identification, drawn dashed. The two therefore enter at different orders, g_6 and g_4 of Eqs. (2) and (3), and only the hexagon survives the source average of Eq. (10). (ii) Heat capacity (solid, left axis) and entropy (dashed, right axis) at $(J_{\pm}, J_{\pm\pm})/J_{zz} = (0.046, 0)$, for ordinary periodic ED (orange) and the winding-free replacement of Eq. (11) at $M = 4$ (blue). Dotted vertical lines mark the perturbative scales g_6 and g_4 of Eqs. (2) and (3); the periodic low-temperature peak tracks the artifact scale g_4 , whereas the projected peak moves down to the ring-exchange scale g_6 . (iii.a),(iii.b) Zero-temperature longitudinal structure factor $S^{zz}(\mathbf{k}, \omega)$ [Eq. (15)] for the periodic band $h(\mathbf{0})$ and the projected band $h_{M=4}$, on identical linear color scales. Both use the same canonically dressed microscopic S^z probe [Eq. (14)]; results are averaged over the three symmetry-related X points, with broadening $\eta = 6 \times 10^{-4} J_{zz}$.

winding four-cycles, 32 contractible hexagons, and 96 wrapping hexagons. Ordinary ED thus probes the parametrically larger scale of a finite-torus winding process rather than the ring exchange governing the emergent gauge sector, and the lower anomaly it produces is centered near g_4 rather than g_6 , as shown in Fig. 1(ii).

In this Letter, we remove this artifact by coupling microscopic spin transfers to an auxiliary source that records their transport on the covering lattice. We map the exact ice-descended band to a fixed reference space and retain only its zero-transport Fourier component. This transport-character projection removes noncontractible winding processes while preserving contractible hexagon dynamics and microscopic virtual dressing. On

cubic-16, the projected band is converged to sub-percent accuracy with source-grid size and recovers the correct low-energy thermodynamic scale at a sign-free benchmark point. The projection also strongly reorganizes the longitudinal spectrum, demonstrating that the winding artifact affects both energies and matrix elements.

Transport-character projection.—The physical and spurious processes can be separated without reference to perturbative order. A microscopic sequence may return to the same spin configuration on the finite torus while transporting Ising polarization through a periodic boundary: it is closed on the torus, but fails to close when lifted continuously to the infinite covering lattice. The winding four-cycles transport nonzero polarization, whereas a

contractible hexagon transports none [Figs. 1(i.a)–1(i.f)]. Equivalently, each exchange creates or moves a spinon–antispinon pair, and the sequence returns to the ice manifold only by annihilating it: on a contractible loop the two charges genuinely meet, whereas on a winding loop each meets a periodic image of its partner, which is why the process survives on the torus but has no counterpart on the covering lattice. We now promote this observation to an exact operator identity.

A periodic cluster is specified by three period vectors, collected as the rows of a 3×3 matrix L , and each site i is assigned a representative position $\mathbf{r}_i$ in a chosen fundamental cell. For every stored oriented nearest-neighbor bond (i, j) we choose $\mathbf{n}_{ij} \in \mathbb{Z}^3$ such that the covering-lattice image of j connected directly to i lies at $\mathbf{R}_{j|i} = \mathbf{r}_j - \mathbf{n}_{ij}^T L$; thus $\mathbf{n}_{ij} = 0$ for an intracell bond, whereas $\mathbf{n}_{ij} \neq 0$ records the periodic image reached by a boundary-crossing bond. The covering-lattice displacement along the oriented bond and the corresponding pair coordinate are $\mathbf{d}_{ij} = \mathbf{R}_{j|i} - \mathbf{r}_i$ and $\mathbf{c}_{ij} = \mathbf{r}_i + \mathbf{R}_{j|i}$.

To quantify transport, consider a spin configuration $\mathcal{C}$ lifted from the torus to the covering lattice, and associate with it the polarization coordinate

$$\mathbf{p}(\mathcal{C}) = \sum_{i=1}^N S_i^z(\mathcal{C}) \mathbf{R}_i, \quad (4)$$

where i runs over all N cluster sites and $\mathbf{R}_i$ is the covering-lattice position assigned to site i by the lift. Equation (4) is the first spatial moment of the site-resolved Ising polarization. It introduces no new physical degree of freedom and is neither an electric nor a magnetic dipole moment, nor a dipole formed from the tetrahedral charges Q_t ; it is a bookkeeping coordinate recording how S^z is displaced by a sequence of microscopic spin flips.

Taken alone, $\mathbf{p}$ is not single valued on a torus: replacing a site by one of its periodic images shifts it by a period vector. Its change along a continuously lifted sequence is unambiguous, however, and for an ice-to-ice sequence is also independent of the coordinate origin, because the ice rule implies $\sum_i S_i^z = 0$. The increments then follow directly from Eq. (4): $S_i^+ S_j^-$ moves one unit of S^z from $\mathbf{R}_{j|i}$ to $\mathbf{r}_i$, giving $\Delta \mathbf{p} = -\mathbf{d}_{ij}$, whereas $S_i^+ S_j^+$ raises both endpoints and displaces $\mathbf{p}$ by their summed position $\mathbf{c}_{ij}$. A move T_ℓ on the oriented bond (i, j) therefore contributes

$$\Delta \mathbf{p}_\ell = \begin{cases} \mp \mathbf{d}_{ij}, & T_\ell = S_i^\pm S_j^\mp, \\ \pm \mathbf{c}_{ij}, & T_\ell = S_i^\pm S_j^\pm. \end{cases} \quad (5)$$

We introduce an auxiliary counting field $\boldsymbol{\theta}$ conjugate to these increments by multiplying each microscopic move

by $e^{-2i\boldsymbol{\theta} \cdot \Delta \mathbf{p}_\ell}$, giving the sourced Hamiltonian

$$H(\boldsymbol{\theta}) = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_\pm \sum_{\langle ij \rangle} (e^{2i\boldsymbol{\theta} \cdot \mathbf{d}_{ij}} S_i^+ S_j^- + \text{h.c.}) + J_{\pm\pm} \sum_{\langle ij \rangle} (e^{-2i\boldsymbol{\theta} \cdot \mathbf{c}_{ij}} S_i^+ S_j^+ + \text{h.c.}). \quad (6)$$

The field $\boldsymbol{\theta}$ is not a physical electromagnetic field; it is a Fourier variable that records the polarization transported by microscopic spin transfers. For any sequence γ the source phases multiply to exactly $e^{-i\boldsymbol{\theta} \cdot \mathbf{q}_\gamma}$, with

$$\mathbf{q}_\gamma = 2 \sum_{\ell \in \gamma} \Delta \mathbf{p}_\ell, \quad (7)$$

the factor of two expressing the transport in the natural normalization $\sigma^z = 2S^z = \pm 1$. The vector $\mathbf{q}_\gamma$ is thus the net Ising-polarization transport of the lifted sequence.

Crucially, $\mathbf{q}_\gamma$ is a property of the path through configuration space, not merely of its endpoints on the finite torus. For a contractible closed process the lifted configuration also closes, the polarization returns to the same branch, and $\mathbf{q}_\gamma = 0$. A winding process instead returns to precisely the same finite-cluster configuration while transferring polarization through a periodic boundary; its lift ends on a different branch and $\mathbf{q}_\gamma \neq 0$. Equation (7) is therefore an exact microscopic winding label, valid beyond perturbation theory and in the presence of both $J_\pm$ and $J_{\pm\pm}$.

Let P_θ be the spectral projector onto the isolated band adiabatically connected to the spin-ice manifold,

$$P_\theta = \frac{1}{2\pi i} \oint_{\Gamma_\theta} (z - H(\boldsymbol{\theta}))^{-1} dz, \quad \text{rank } P_\theta = \text{rank } P_{\text{ice}}, \quad (8)$$

where the contour Γ_θ encloses the selected band and no other part of the spectrum. Because the band eigenvectors depend on $\boldsymbol{\theta}$, their matrix representations cannot be averaged directly. We therefore identify every sourced band with the same fixed ice-rule reference space using the polar isometry W_θ , and define the exact band Hamiltonian in that common basis,

$$W_\theta = P_\theta P_{\text{ice}} (P_{\text{ice}} P_\theta P_{\text{ice}})^{-1/2}, \quad h(\boldsymbol{\theta}) = W_\theta^\dagger H(\boldsymbol{\theta}) W_\theta. \quad (9)$$

A continuous source average retains only the zero-transport Fourier component, which we evaluate numerically on an M^3 grid,

$$h_{\text{wf}} = \frac{1}{(2\pi)^3} \int_{[0, 2\pi]^3} d^3 \boldsymbol{\theta} h(\boldsymbol{\theta}) \simeq h_M = \frac{1}{M^3} \sum_{\mathbf{m} \in \mathbb{Z}_M^3} h\left(\frac{2\pi \mathbf{m}}{M}\right). \quad (10)$$

The continuous average selects $\mathbf{q}_\gamma = 0$ exactly, removing every process that carries nonzero covering-lattice transport; the discrete grid instead retains all characters with $\mathbf{q}_\gamma = \mathbf{0} \pmod{M}$. A sequence of n exchanges crosses the boundary at most n times, so $\|\mathbf{q}_\gamma\|_\infty \leq n$ and a nonzero

character can survive an M^3 grid only at order $n \geq M$. Finite M is therefore exact through order $M - 1$, with a leading alias of known onset rather than an uncontrolled truncation. One consequence of the continuous average is available exactly at any M : since $\mathbf{q}_\gamma = 2\Delta\mathbf{p}$, a surviving matrix element cannot change $\mathbf{p}$, so h_{wf} is block diagonal in the polarization coordinate of Eq. (4). We therefore impose that block structure directly on h_M . The residual inter-sector coupling it discards is only $1.6 \times 10^{-5} J_{zz}$ at $M = 4$, but it is comparable to the splittings within the low-lying manifold and would otherwise lift degeneracies that the exact projection protects. The construction ceases to be well defined if the selected band closes its spectral gap or if $P_{\text{ice}} P_\theta P_{\text{ice}}$ becomes singular.

For thermodynamics we replace only the exact band descended from the ice manifold,

$$Z_{\text{wf}}(\beta) = Z_{\text{full}}(\beta) - \text{Tr}_{\text{ice}} e^{-\beta h(\mathbf{0})} + \text{Tr}_{\text{ice}} e^{-\beta \bar{h}_M}, \quad (11)$$

and perform the analogous substitution for the first two energy moments that give the internal energy and heat capacity; the microscopic spectrum outside the selected band is left unchanged. Here

$$\bar{h}_M = h_M + \frac{\text{Tr}[h(\mathbf{0}) - h_M]}{d} I, \quad d = \text{rank } P_{\text{ice}}, \quad (12)$$

fixes the scalar placement of the corrected band relative to the microscopic spectrum. The band Hamiltonians of Eq. (9) are referred to the ice manifold and therefore carry no absolute energy origin, which the character average of Eq. (10) does not supply; matching the band trace to $h(\mathbf{0})$ supplies it. Because a uniform shift of an isolated band cancels in both C and S , this choice leaves the reported low-temperature observables unchanged. Equation (11) defines a finite-size diagnostic rather than the partition function of the original periodic Hamiltonian. Derivations and numerical details are given in the Supplemental Material [13].

Validation against perturbation theory.—The projection is defined nonperturbatively, but wherever the loop expansion is controlled the projected band must reduce to pure hexagon ring exchange. This is a quantitative test, because the band Hamiltonian is written in the fixed reference space spanned by the $d = 90$ zero-spinon configurations of cubic-16: the matrix element between two configurations that differ by the spins of a given loop is that loop's amplitude, and can be read off and compared with Eqs. (2) and (3). We do this at the sign-free benchmark point

$$(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0), \quad (13)$$

where $g_6 = 1.168 \times 10^{-3} J_{zz}$ and $g_4 = 8.464 \times 10^{-3} J_{zz}$.

In the periodic band $h(\mathbf{0})$ each of the 36 winding four-loops carries $1.046 \times 10^{-2} J_{zz} = 1.24 g_4$, exceeding the contractible hexagon amplitude $1.314 \times 10^{-3} J_{zz} = 1.13 g_6$

by a factor of 8.0: the leading ice-to-ice process of ordinary ED is the artifact, not the ring exchange. In the projected band $h_{M=4}$ all 36 winding four-loop amplitudes and all 48 wrapping-hexagon amplitudes vanish, $|h_{ij}| \leq 1.2 \times 10^{-16} J_{zz}$, fourteen orders of magnitude below g_4 and at the level of the arithmetic. All 16 contractible hexagon amplitudes survive at $1.325 \times 10^{-3} J_{zz} = 1.13 g_6$, differing from their periodic values by 0.8%. Their 13% excess over bare third order is higher-order virtual dressing, which the exact-band construction retains by design. Repeating at $J_\pm/J_{zz} = 0.030$ assigns the two channels without perturbative input: they scale as $J_\pm^{2.13}$ and $J_\pm^{3.01}$, matching g_4 and g_6 . The same audit on FCC-32 removes all 24 winding four-loops and all 96 wrapping hexagons while retaining all 32 contractible hexagons. The band is well isolated throughout, and the retention census, band gap, and pullback diagnostics are given in the Supplemental Material [13].

The source grid needed to remove the artifact is not a matter of convergence. Enumerating every ice-to-ice sequence through third order, on both cubic-16 and FCC-32, we find that all transport characters have components in $\{0, \pm 1\}$, saturating the bound below Eq. (10) at $n = 1$ rather than $n = 3$. A character with unit components is never $\mathbf{0} \bmod 2$, so the $M = 2$ grid already annihilates every winding process available at the orders where the artifact lives, exactly and with no extrapolation: the four-loop amplitudes above are $10^{-19} J_{zz}$ at $M = 2$. Larger M addresses only higher-order sequences whose transport is commensurate with the grid, and the residual M dependence measures them: the centered band operator changes by 6.36% from $M = 2$ to $M = 3$ and 0.013% from $M = 3$ to $M = 4$, and the winding amplitude that the odd grid $M = 3$ retains is a sixth-order alias, scaling as $J_\pm^{6.25}$. We quote $M = 4$ throughout as a margin against such higher-order aliasing, but the removal of the artifact itself requires only $M = 2$.

Thermodynamics and dynamics.—With the artifact removed we can compare against QMC. At the benchmark point of Eq. (13), continuous-time worm QMC on 8^3 primitive cells [10] places the low-temperature heat-capacity maximum at $T/J_{zz} \approx 0.001100$. Ordinary cubic-16 ED gives 0.02554, tracking the four-loop scale g_4 ; the winding-free replacement gives 0.001463, tracking g_6 [Fig. 1(ii)]. Over the low-temperature window the heat-capacity root-mean-square error drops from 0.0859 to 0.02327, while the entropy error is essentially unchanged at 0.04509. The projection thus fixes the low-energy scale; it does not make a 16-site torus a quantitative model of the bulk.

The same comparison can be made in the spectrum. To probe the periodic and projected bands with an identical microscopic operator, we map the spin operator into the fixed ice reference space with the zero-source isometry

W_0 , defining for sublattice a

$$\tilde{S}_a^z(\mathbf{k}) = W_0^\dagger \left(\sum_{j \in a} e^{-2\pi i \mathbf{k} \cdot \mathbf{r}_j} S_j^z \right) W_0, \quad (14)$$

where $\mathbf{k}$ is the reciprocal-space momentum, distinct from the transport label $\mathbf{q}_\gamma$. For either the periodic band Hamiltonian $h(\mathbf{0})$ or the projected Hamiltonian $h_{M=4}$, the zero-temperature longitudinal structure factor is

$$S^{zz}(\mathbf{k}, \omega) = \frac{1}{N d_0} \sum_{g \in \mathcal{G}} \sum_{n \notin \mathcal{G}} \sum_{a=0}^3 \left| \langle n | \tilde{S}_a^z(\mathbf{k}) | g \rangle \right|^2 \times G_\eta [\omega - (E_n - E_g)], \quad (15)$$

where $\mathcal{G}$ is the ground-state manifold, $d_0 = |\mathcal{G}|$, N is the number of sites, $\eta = 6 \times 10^{-4} J_{zz}$ is the spectral broadening, and $|g\rangle$, $|n\rangle$ are eigenstates of the band Hamiltonian under consideration.

Cubic-16 resolves only Γ and three symmetry-related X points. At X , the spectral peak shifts from $0.08069 J_{zz}$ in the periodic band to $0.005362 J_{zz}$ after projection [Figs. 1(iii.a) and 1(iii.b)]. Since X sits on the zone boundary, this is directly comparable to the photon bandwidth $W_\gamma \approx 5 \times 10^{-3} J_{zz}$ extracted from QMC with stochastic analytic continuation at the same coupling [10]: the projected peak agrees to 7%, while the periodic band overshoots sixteenfold. At Γ the projected response vanishes identically, because the $\mathbf{k} = \mathbf{0}$ probe is the sublattice magnetization, a c-number on each polarization block; the QMC-SAC spectrum likewise carries all of its $\mathbf{q} = \mathbf{0}$ weight at zero energy [10]. The projection therefore reorganizes excitation energies and spectral weights together, rather than rescaling the periodic spectrum.

One artifact of exact polarization conservation should be noted: the projected ground state is sixfold degenerate, the sectors $\mathbf{p} = (\pm \frac{1}{2}, 0, 0)$, $(0, \pm \frac{1}{2}, 0)$, $(0, 0, \pm \frac{1}{2})$ being unable to tunnel into one another. The resulting $\ln 6$ per cluster is a floor under the low-temperature entropy on cubic-16, and accounts for part of the residual entropy discrepancy, but the sector count grows only polynomially with linear size and so vanishes per site in the thermodynamic limit.

Discussion.—Periodic ED replaces the physical hexagon dynamics of QSI with a parametrically larger noncontractible four-spin process. Transport-character projection removes it from the exact ice-descended band while preserving contractible processes, microscopic virtual dressing, and the spectrum outside the band. Read directly in the zero-spinon basis, the projected band has no winding amplitude left at the 10^{-16} level and retains every hexagon at the ring-exchange scale; with that established, the corrected cluster reproduces both the QMC heat-capacity scale and the QMC photon bandwidth at the sign-free point. What remains is genuine finite-size error: cubic-16 has no long-wavelength

photon sector and sparse momentum resolution, so the projection is a controlled finite-torus correction, not a substitute for thermodynamic extrapolation.

The method is sign unrestricted, so the natural next step is a character-converged FCC-32 calculation at the π -flux XYZ parameters relevant to $\text{Ce}_2\text{Hf}_2\text{O}_7$, where QMC cannot follow.

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- [1] M. Hermele, M. P. A. Fisher, and L. Balents, Pyrochlore photons: The $U(1)$ spin liquid in a $S = 1/2$ three-dimensional frustrated magnet, *Phys. Rev. B* **69**, 064404 (2004).
- [2] M. J. P. Gingras and P. A. McClarty, Quantum spin ice: A search for gapless quantum spin liquids in pyrochlore magnets, *Rep. Prog. Phys.* **77**, 056501 (2014).
- [3] O. Benton, O. Sikora, and N. Shannon, Seeing the light: Experimental signatures of emergent electromagnetism in a quantum spin ice, *Phys. Rev. B* **86**, 075154 (2012).
- [4] B. Gao, T. Chen, D. W. Tam, C.-L. Huang, K. Sasmal, D. T. Adroja, F. Ye, H. Cao, G. Sala, M. B. Stone, C. Baines, J. A. T. Barker, W. Hu, J.-H. Chung, X. Xu, S.-W. Cheong, M. Nallaiyan, S. Spagna, M. B. Maple, A. H. Nevidomskyy, E. Morosan, G. Chen, and P. Dai, Experimental signatures of a three-dimensional quantum spin liquid in effective spin-1/2 $\text{Ce}_2\text{Zr}_2\text{O}_7$ pyrochlore, *Nat. Phys.* **15**, 1052 (2019).
- [5] J. Gaudet, E. M. Smith, J. Dudemaine, J. Beare, C. R. C. Buhariwalla, N. P. Butch, M. B. Stone, A. I. Kolesnikov, G. Xu, D. R. Yahne, K. A. Ross, C. A. Marjerrison, J. D. Garrett, G. M. Luke, A. D. Bianchi, and B. D. Gaulin, Quantum spin ice dynamics in the dipole-octupole pyrochlore magnet $\text{Ce}_2\text{Zr}_2\text{O}_7$, *Phys. Rev. Lett.* **122**, 187201 (2019).
- [6] A. Bhardwaj, S. Zhang, H. Yan, R. Moessner, A. H. Nevidomskyy, and H. J. Changlani, Sleuthing out exotic quantum spin liquidity in the pyrochlore magnet $\text{Ce}_2\text{Zr}_2\text{O}_7$, *npj Quantum Mater.* **7**, 51 (2022).
- [7] E. M. Smith, A. Fitterman, R. Schaffer, *et al.*, Two-peak heat capacity accounts for $R \ln 2$ entropy and ground state access in the dipole-octupole pyrochlore $\text{Ce}_2\text{Hf}_2\text{O}_7$, *Phys. Rev. Lett.* **135**, 086702 (2025).
- [8] N. Shannon, O. Sikora, F. Pollmann, K. Penc, and P. Fulde, Quantum ice: A quantum monte carlo study, *Phys. Rev. Lett.* **108**, 067204 (2012).
- [9] Y. Kato and S. Onoda, Numerical evidence of quantum melting of spin ice: Quantum-to-classical crossover, *Phys. Rev. Lett.* **115**, 077202 (2015).
- [10] C.-J. Huang, Y. Deng, Y. Wan, and Z. Y. Meng, Dynamics of topological excitations in a model quantum spin ice, *Phys. Rev. Lett.* **120**, 167202 (2018), arXiv:1707.00099.
- [11] R. Schäfer, I. Hagymási, R. Moessner, and D. J. Luitz, Pyrochlore $S = \frac{1}{2}$ heisenberg antiferromagnet at finite temperature, *Phys. Rev. B* **102**, 054408 (2020).
- [12] I. Hagymási, R. Schäfer, R. Moessner, and D. J. Luitz, Possible inversion symmetry breaking in the $S = 1/2$ pyrochlore heisenberg magnet, *Phys. Rev. Lett.* **126**, 117204 (2021).

[13] (2026), see Supplemental Material for the canonical block construction, transported-dipole character projec-

tion, full-Hilbert counterterm, thermodynamic identities, geometry, and benchmark provenance.